3ai	$mg(\Delta)h = 64 \times 9.81 \times 120 = 75\,340.8$ J $= 75$ kJ	A1
3aii	Elastic potential energy = area under F-h graph $= 0.5 \times 1800 \times (120 - 37)$ $= 75\,000$ J $= 75$ kJ	 C1 A1
3b	Gravitational potential energy store has been fully transferred to / equal to elastic potential energy store Kinetic energy store is zero so <u>speed is zero</u>.	B1 B1
4ai	*g* and *r* are constants, so *a* is proportional to *x*.	B1